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Demographic Parameters of Iberian Killer Whales Between 2011 and 2023

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Received: 19 January 2025 | **Revised:** 9 October 2025 | **Accepted:** 10 October 2025

Funding: This work was partially funded by the project FriendSHIP: Orca with funds from the Fundación Banco Santander (https://www.orcaiberica.org/; App: GT ORCAS).

Keywords: abundance | demographic parameters | killer whale | *Orcinus orca* | survival rate

ABSTRACT

The Iberian killer whale subpopulation was assessed as critically endangered (CR) by the IUCN red list of threatened species based on the available information until 2011 on its low abundance, high newborn mortality, and dependency on the endangered Atlantic bluefin tuna. The present study focuses on updating the subpopulation demographic parameters between 2011 and 2023. Data were based on photo-identification and stranding records. Abundance, survival, reproductive rates, and population growth were estimated through mark-recapture models. Stranding records were used to estimate body size and calving seasonality. Sightings of newborn calves were also used for assessing seasonality. From 2011 through 2023, 26 stranding records were obtained, and 18,554 ID-images were examined. This subpopulation showed a stable abundance, with 37 individuals in 2023, still qualifying for the IUCN CR category under criteria D. Calf survival rate has increased compared to pre-2011, while adult survival has decreased, particularly for females. Reproductive rates were low, with an estimated interbirth interval of 8.3 years. The overall population growth rate was apparently stable at 0.46%. Most births occurred in the summer and fall. Iberian killer whales are generally smaller than the average body length of other killer whale populations throughout the world. The study emphasizes the need for long-term monitoring and highlights new potential threats from fishing activities and recent disruptive interactions with recreational boats. The future of the subpopulation depends on recruitment and survival of juveniles and effective female reproduction.

1 | Introduction

The killer whales in the Iberian Peninsula are isolated from other subpopulations in the Northeast Atlantic, according to research on photo-identification, mitochondrial DNA, and microsatellite genetic markers, and other ecological biomarkers (Esteban, Verborgh, Gauffier, Alarcón, et al. 2016; Esteban, Verborgh, Gauffier, Giménez, Martín, et al. 2016; Foote et al. 2011). Their demographic parameters were estimated at very low levels between 1999 and 2011, with a high newborn mortality observed at the end of the period and a small number of individuals forming the subpopulation (Esteban, Verborgh,

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Gauffier, Giménez, Guinet, et al. 2016). They are known to feed predominantly on Atlantic bluefin tuna (*Thunnus thynnus*) (García-Tiscar 2009) and exhibit different foraging strategies, including actively chasing medium-sized tuna (Guinet et al. 2007) or depredating tuna from fishing lines in the Strait of Gibraltar (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016). This depredation strategy has been observed only among some social groups (Esteban, Verborgh, Gauffier, Giménez, Foote, et al. 2016), which exhibit higher survival and reproductive outputs than non-depredating social groups (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016). From 2011 to 2015, a lower newborn mortality was observed in the Strait of Gibraltar (Otero-Sabio et al. 2018). Since then, no new estimate is available.

The subpopulation was declared as critically endangered by the International Union for the Conservation of Nature (IUCN) in 2019 under criteria C2a(i,ii) as there were far fewer than 50 mature individuals and criteria D because the whales were heavily dependent on an endangered prey species (bluefin tuna) (Esteban and Foote 2019). Furthermore, a continuing decline in the number of mature individuals was projected for the future, based on their poor recruitment (Esteban and Foote 2019). Based on this assessment, the subpopulation has been included as critically endangered in the Portuguese Red List, since orcas that occur in waters off mainland Portugal are part of this subpopulation (Mathias et al. 2023). The Iberian killer whale subpopulation is recognized as endangered or vulnerable in its range countries and is protected under national and international legislation.[1]

The conservation status of the Iberian killer whale subpopulation is linked to that of their main prey, Atlantic bluefin tuna (Esteban, Verborgh, Gauffier, Alarcón, et al. 2016; Esteban and Foote 2019; García-Tiscar 2009). Bluefin tuna is an important species in both commercial and recreational fisheries (McKinney et al. 2020). The species was largely overfished during the 1990s to early 2000s, decreasing its stock biomass, which led to worries about its collapse (Fromentin and Powers 2005; MacKenzie et al. 2009). The bluefin tuna fishery is managed by the International Commission for the Conservation of Atlantic Tunas (ICCAT), which unveiled a 15-year recovery plan in 2007 (ICCAT 2006). The decrease of fishing quotas for ICCAT member states resulted in the recovery of the population (ICCAT 2017). A management plan that moved from recovery to sustainable exploitation was adopted by ICCAT in 2018 (ICCAT 2018). The IUCN officially recognized bluefin tuna's recovery, reassessing it to "Least Concern" in 2021 (Collette et al. 2021). It might then be expected that the positive status of Iberian killer whale's main prey in recent years could improve killer whales' survival and reproduction and contribute to the recovery of this critically endangered subpopulation.

However, new threats have emerged for this small subpopulation. Tuna fishers have started using new fishing techniques in the Strait of Gibraltar, and a new disruptive behavior has spread among the killer whales. Since 2020, interactions between killer whales and vessels have been registered in the Atlantic waters of the Iberian Peninsula, where the animals approach vessels with unclear motivation and often strike and turn the rudders, causing severe hull damage in some cases (Esteban et al. 2022). Since the first described occurrence in May 2020, these interactions have increased rapidly both in frequency and geographic extension. Esteban et al. (2022) found that medium-sized sailboats with spade rudders were targeted most frequently by killer whales. Between 2020 and 2021, 31 killer whales were identified, of which nine individuals interacted with vessels (Esteban et al. 2022, 2023). An updated report identified up to 14 individuals from 4 groups interacting with vessels (Esteban et al. 2023). This situation has caused great public concern, especially in the recreational sailing community, as this multifaceted issue can put both whale and human lives at risk of injury or worse. This unusual behavior is still occurring as of October 2025.

The aim of this study is to estimate demographic parameters of the Iberian killer whales between 2011 and 2023 with opportunistic data, which is the only data source during this period. This is of particular importance given the endangered status of this subpopulation and the recent disruptive behavior of the whales, which might imperil their already fragile status.

2 | Methods

In this study, we will refer to *interaction* when killer whales (interacting and non-interacting animals) approach or make physical contact with vessels, mainly recreational sailing vessels, as described by Esteban et al. (2022, 2023). Meanwhile, *depredation* will refer to killer whales (depredating and non-depredating animals) approaching or making physical contact with fishing lines, as described by Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. (2016).

2.1 | Data-Collection

Opportunistic photographs, useful for photo-identification, were obtained mostly from crew onboard whale-watching companies operating in the Strait of Gibraltar, Portugal, and northern Spain between 1999 and 2023. We also received photos taken by naturalists, captains, professional photographers, and vessel-based surveys conducted by several NGOs or research units such as CEMMA, CSIC-IEO, AMBAR, and AZTI in northern Spain and the Bay of Biscay. All photographs were taken opportunistically without any predefined protocol. Contributions from citizen scientists to this catalogue were encouraged via public presentations, local contacts, and, more recently, social media.

Distinct natural markings, scars, and pigmentation patterns can be used to identify individual killer whales through photo-identification (e.g., Bigg 1987). Photo-identification techniques were used as previously described in Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. (2016). Photographs of the killer whales' saddle, eye patches, and dorsal fin were taken from both sides whenever the whales were spotted. Whales that were identified were assigned a unique alpha-numeric code in the catalogue. Matches to previously identified individuals were made by comparing each new photograph with every other photograph in the photo identification collection. The quality of the photos was rated as follows: poor (Q0), good (Q1), and excellent (Q2: flawlessly focused photo taken at a perpendicular angle with the entire fin exposed). To reduce the possibility of misidentification, only images Q1 and Q2 were used in this study.

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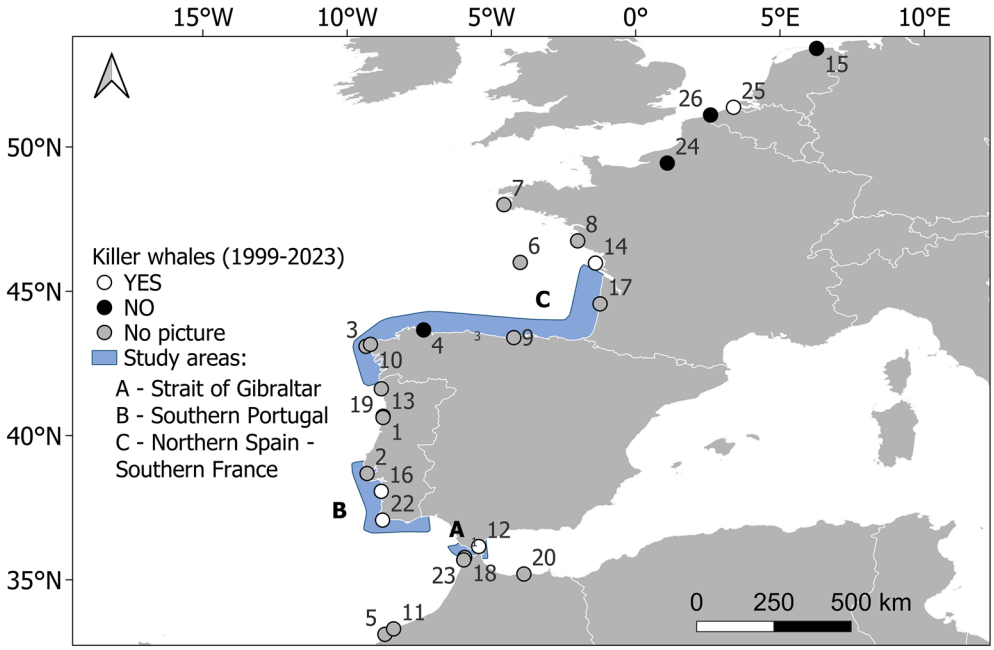


FIGURE 1 | Study areas along the Iberian Peninsula (blue areas). Points indicate the location of strandings, including whether there was a match with the photo-ID catalogue of the Iberian killer whales (YES), or not (NO); or whether no good-quality pictures of their distinguishing features (such as the trailing edge of dorsal fin, eye patch or saddle patch) was taken, or the animal was too decomposed to be identified (No picture). Numbers near the points correspond to strandings listed in Table 1 ordered by stranding date.

Social structure analysis is out of the scope of this manuscript; therefore, animals seen together over several years or based on known mother-offspring relationships were considered as belonging to the same community but are not directly comparable with the social groups defined in Esteban, Verborgh, Gauffier, Giménez, Foote, et al. (2016). A community was considered as belonging to the Iberian subpopulation when its individuals were observed in several locations along the Iberian Peninsula (at least areas 1 and 2 of Figure 1) in at least three consecutive years. Each individual was then assigned a unique identification (ID) number composed of the letter of its community followed by three unique digits within each community.

In order to understand long-distance movements and links to other areas, this catalogue was compared with other existing catalogues in the North Atlantic Ocean.

Killer whale stranding records were compiled from published and unpublished sources from the Atlantic coast of Morocco, Spain, Portugal, and France along their known distribution (Esteban et al. 2022; Pouey-Santalou et al. 2024), and the nearby area comprising the Mediterranean Sea, Belgium, and the Netherlands.

2.2 | Demographic Parameters

2.2.1 | Abundance and Calving Season

According to Esteban, Verborgh, Gauffier, Giménez, Foote, et al. (2016), Iberian killer whales probably have a matrilineal social structure in which individuals do not disperse from their natal group. As a result, if a whale was discovered stranded or had not been seen with its pod for 3 years in a row, it was deemed dead or having permanently emigrated. An annual cumulative abundance count was therefore computed by subtracting the number of individuals considered deceased or having permanently emigrated from the total number of individuals identified, based on annual censuses. We estimated the age and sex of the whales using photographs following Olesiuk et al. (1990). We divided the whales into four classes based on the criteria for sexual maturity given by Olesiuk et al. (1990). Mature males with a conspicuous dorsal fin and mature females with calves or juveniles in echelon formation were considered adults; juveniles were those who were older than a year but had not reached adulthood, and calves were those who were younger than a year. Visitors were defined as individuals who had been sighted less than five times and in only one of the study areas (Figure 1) and were not used to compute the cumulative abundance of the Iberian killer whale subpopulation.

Information on the seasonality of calving in the study area was compiled from two source types: (1) calf strandings and (2) the appearance of newborn calves between consecutive encounters.

2.2.2 | Survival, Reproductive and Population Growth Rates

Mark-recapture analysis was done over a subset of the data as effort has been consistent over these years, areas and months: (i) the period considered was 2011–2023 (hereafter, the modeled period) to expand on previous studies from 1999 to 2011 (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016; Ouled-Cheikh et al. 2023), (ii) the study areas were the Strait of Gibraltar and southern Portugal (areas A and B in Figure 1) and (iii) the data

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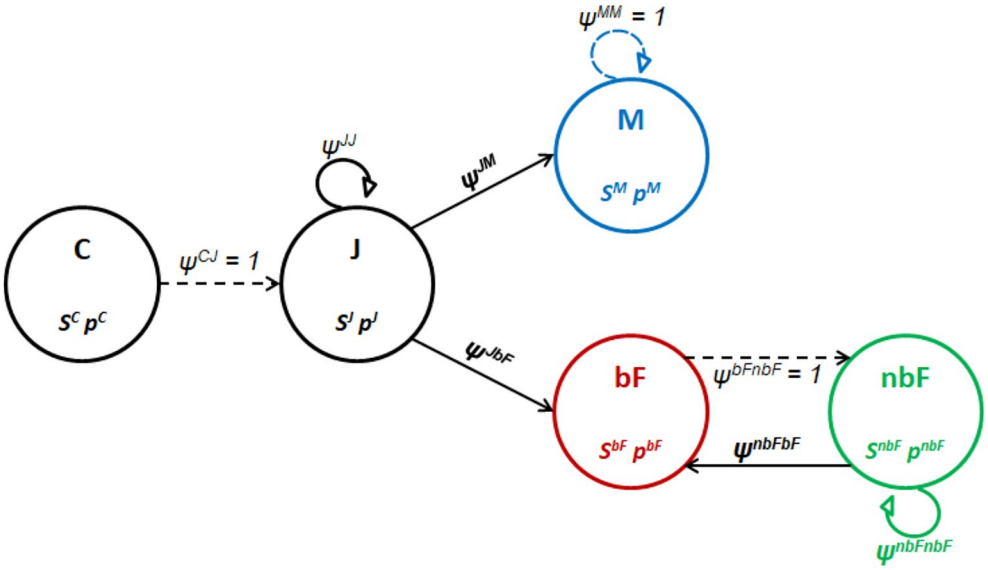


FIGURE 2 | Demographic parameters estimated by the multistate model for survival rate S^r, recapture probability p^r, and transition probabilities ψ^{rs} for the five reproductive classes: Calf (C), juvenile (J), adult male (M), breeding female (bF), and non-breeding female (nbF). For clarity, "impossible" transitions, that is, set to 0, are not shown on the figure. Transitions set to 1 are represented by a broken line, and transition probabilities estimated by the model are shown as continuous lines.

were restricted between May and October. Therefore, the analyses were restricted to whales from the Iberian subpopulation. The year 2011 was included in our study to estimate the survival of calves born in 2011. Multi-state mark-recapture models (Brownie et al. 1993; Hestbeck et al. 1991) were used to estimate reproduction rates and survival rates of different age classes. Data from May to October were pooled annually (= an occasion t).

The following assumptions applied to the mark-recapture models used (Pradel 1996). (1) Animals are marked and released immediately; in the case of photo-identification, no physical capture takes place and therefore the animals are never taken out of the study area for marking. (2) No mark is lost or misread; each individual was followed for new marks that could appear, and matching was confirmed by at least 2 experts. (3) There is no temporary emigration (except possibly random); data were tested for overdispersion (transience and trap-dependence) through a goodness-of-fit (GOF) test for multistate and CJS models in the program U-CARE (Choquet et al. 2009). (4) Animals are uniquely identifiable; only high-quality photographs were used to positively identify each individual. (5) Animals have independent fates; although fate independence cannot be ascertained in the case of the death of a mother, its calf is unlikely to survive; the fact that each age class was modeled separately allows meeting this assumption.

The following parameters were estimated:

- Apparent annual survival rate (called survival rate hereafter), $S^r{}_t$, is the probability that an individual survives from occasion t to occasion $t+1$ given that it is in state r at occasion t. Survival depends only on the individual's state at occasion t.
- Transition probability, $\psi^{rs}{}_t$, the conditional probability that an animal in state r at occasion t is in state s at occasion $t+1$, given that the animal is alive at $t+1$. The transition step from state r to state s can only happen after the animal has already survived to period $t+1$. On any occasion, the sum of all transition probabilities $\psi^{rs}{}_t$ from all states r to any given state s equals 1.

2.3 | Model With Five Reproductive States

In the first model, states r represented the following reproductive states for each individual at occasion t: (C) newborn calf [0–1 year], (J) juvenile, (M) adult male, (bF) breeding female, that is, adult female seen with a newborn calf, and (nbF) non-breeding female = adult female seen without a newborn calf. Impossible transitions were fixed to zero, that is, adults could not become juveniles or calves, and females could not become males and vice versa. Because the transition probability between calf and juvenile was constant, that is, a calf becomes a juvenile after 1 year, this parameter was fixed at 1 (Figure 2). Adult females cannot have a newborn calf 2 years in a row, so the transition probability from breeding female to non-breeding female was also fixed at 1. Only three transition probabilities were estimated by the model: the annual transition probability from juvenile to breeding female or to adult male provided the average time (the inverse of the rate) an individual is considered juvenile in the photo-identification data set. Additionally, the transition from non-breeding female to breeding female estimated the reproductive rate for adult female whales. The inverse of this rate represents the inter-birth interval (IBI). Note that although, biologically, juveniles can become non-reproductive adult females, it was not possible to evaluate those transition probabilities from our dataset. Since juveniles can have similar sizes as adult females, an individual would therefore be assigned a juvenile state until it is either observed as an adult male, that is, with a large dorsal fin, or as an adult reproductive female, that is, observed with a dependent calf. Since transitions from juvenile to adult states happen only once in the history of each individual, the time-dependent model for these parameters was not informative and was therefore only modeled as constant.

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- The recapture probability p^r was also estimated. Since calves could not be recaptured in the same state, their recapture probability was fixed at 0.

The initial model was $\{S^C(t)\ S^J(t)\ S^{bF}(t)\ S^{nbF}(t)\ S^M(t)\ p^C(.)\ p^J(t)\ p^{bF}(t)\ pn^{bF}(t)\ p^M(t)\ \psi^{JbF}(.)\ \psi^{JM}(.)\ \psi^{nbFbF}(t)\}$, where (t) stands for time-dependent parameters and $(.)$ stands for constant parameters over the study period.

From this initial model, different hypotheses were tested by steps. In step I, the recapture probabilities were set as constant $p^r(.)$ or equal between states when estimates were similar or if it would help the model to estimate the parameters. In step II, the transition probabilities were set as constant $\psi^{rs}(.)$. In step III, survival rates were set as constant $S^r(.)$ or equal between states when estimates were similar or if it would help the model to estimate the parameters. All mark-recapture analyses were performed in MARK 9.0 (White and Burnham 1999). Models were compared and selected using the quasi-likelihood adjusted Akaike Information Criterion adjusted for small sample sizes (QAICc; Cooch and White 2009; Sugiura 1978):

$$\text{QAICc} = \frac{-2ln(L)}{\hat{c}} + 2K + \left(\frac{2K(K+1)}{n-K-1}\right)$$

where L is the likelihood of the model, $\hat{c}$ is the measure of the lack of fit between the general and saturated models, K is the number of parameters, and n the sample size.

The difference between QAICc and any given model and the best supported model (ΔQAICc) was used to assess the relative fit of the model. The best model of each step, based on a ΔQAICc ≤ 2, was selected to test more hypotheses at the next step. At the end, ΔQAICc models ≤ 2 were considered to have a similar fit to the data (Burnham and Anderson 1998).

2.4 | Model for Reproductive Females Only

Another multi-state model was built using only data of adult females that were observed with a calf during the study period (2011–2023) in order to minimize the impact of older females in menopause and to obtain a lower limit estimate of the IBI. This model may, however, exclude females that had calves during the study period that did not survive long enough for them to be observed with their mother (i.e., observed as a breeding female state, bF). Therefore, the estimate will be similar to a viable calf interval. For this model, there were only two states: breeding (bF) and non-breeding (nbF) female, both with the same characteristic given before. The same modeling strategy described before was followed; however, the survival rates were modeled in the second step and the transition rates in the third step.

2.5 | Recruitment and Population Growth Rate Models

Finally, two simple models estimating recruitment (f) and population growth rate, lambda (λ), together with survival (φ) and recapture probability (p) (Pradel 1996), were applied to the Iberian killer whale dataset without states. For both models, all parameters were set as constant in order to obtain overall recruitment and population growth rates.

3 | Results

3.1 | Photo-Identification Catalogue

A total of 21,889 pictures were analyzed from 1999 to 2023. Photographic effort varied annually and increased in the modeled period (since 2011; Figure 3 and S3). Despite this increased effort, the cumulative curve of individuals identified was stable over the modeled period (Figure 3). Since 1999, 71 individuals were identified and included in the catalogue, available at this link. Of these, 21 individuals, including 6 calves, 7 juveniles, 2 adult males, and 6 adult females, were considered dead over the study period. Seventeen of these deaths happened in the modeled period of 2011–2023; only three of those individuals were confirmed dead from stranding records (Table 1). Considering the other two individuals confirmed dead in 2009 and 2015 (Table 1), this accounts for 19% of the stranded animals in their distributional range (Figure S1) that have been confirmed to belong to this subpopulation. There were 21 other known strandings of killer whales along their distributional range during the same study period (1999 to 2023). Three of those strandings were individuals not included in our catalogue (12%), while for the rest ($n = 18$; 69%), either the state of decomposition of the animals did not allow their identification, or there were no pictures of their identifiable features (Table 1).

One individual was observed in 2011 with a LIMPET tag on the base of its dorsal fin. After the tag fell off, one of the attachment darts stayed in the animal until at least July 2013. During this period, the animal was observed on several occasions with linear lacerations behind the dart, probably resulting from depredating on tuna drop-lines. This individual has never been observed within its group since 2016 and was therefore considered dead for the purpose of our study.

Based on the criteria set for this study, out of the 71 individuals in the catalogue, 57 individuals distributed over six communities are considered to belong to the Iberian killer whale subpopulation. Four of these communities have been observed in all the studied areas; the other two only in the southern Iberian Peninsula. The rest of the 14 individuals included in the catalogue are considered visitors, as they have been observed on fewer than five occasions and in only one of the studied areas.

3.2 | Comparison With Other Areas

Two visitor individuals, an adult and a juvenile, which have only been seen in Algarve in 2019 and 2022, were matched with the catalogue from the Azores ($n = 63$; Cabayol 2022).

There were no matches with the catalogues of Scotland ($n = 193$; Scullion et al. 2021), Norway ($n = 971$; Jourdain and Karoliussen 2018), Iceland ($n = 567$; Samarra et al. 2017), Arctic Canada ($n = 63$; Kucheravy et al. 2023), Caribbean ($n = 35$;

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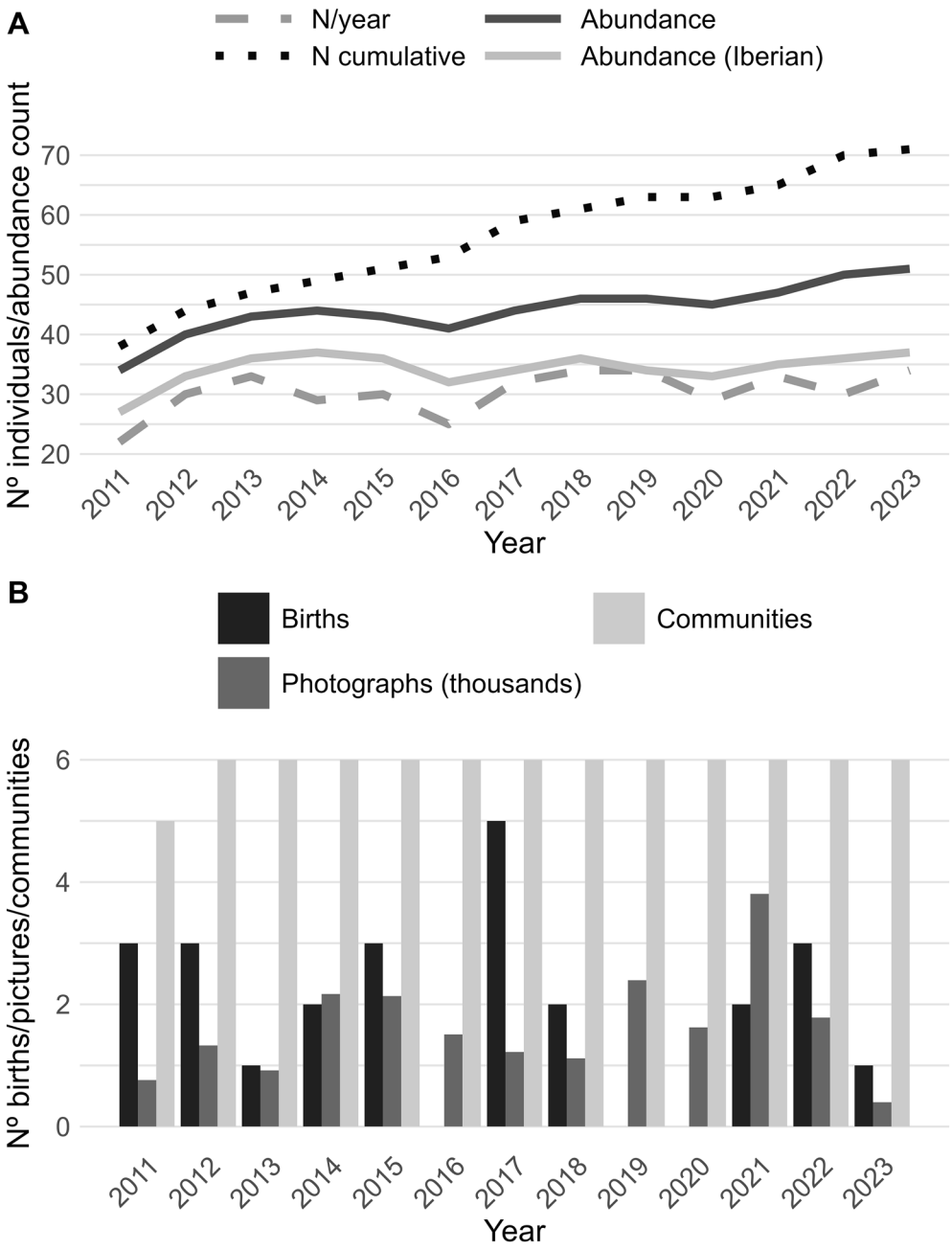


FIGURE 3 | Summary of killer whales identified between 2011 and 2023. (A) The dashed gray line indicates the number of individuals observed per year (*N*/year), the dotted black line indicates the cumulative number of individuals observed over the study period (*N* cumulative), the solid dark gray line indicates the total abundance count (Abundance—total), and the solid light gray line indicates the cumulative Iberian subpopulation abundance count (Abundance—Iberian). Abundance counts were computed by subtracting the number of individuals considered deceased or having permanently emigrated from the total number of individuals observed. (B) Dark gray bars indicate the annual number of photographs analyzed in thousand, black bars indicate the annual number of births and light gray bars indicate the cumulative number of communities identified over the study period.

Bolaños-Jiménez 2023), Canary Islands ($n = 43$; Esteban pers. comm.) or Madeira ($n = 32$; Esteban pers. comm.).

3.3 | Abundance

All communities of Iberian killer whales were identified in 2012. Since then, newly identified individuals were calves from known females of the existing communities or individuals that have been seen sporadically out of the Strait of Gibraltar (Figure 3). Consequently, we considered that the first complete abundance count was in 2012 with 40 individuals, increasing to 51 in 2023. For the Iberian subpopulation, the abundance count was 33 individuals in 2012, remaining fairly constant with 37 individuals in 2023 (Figure 4).

No genetic information is available for the visitors, so the connection of those individuals to the Iberian subpopulation remains unknown. Hereafter, we will only focus on the Iberian killer whale subpopulation, that is, excluding individuals considered as visitors.

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TABLE 1 | Compilation of killer whale strandings (2000–2023) along their known distributional range (Figure S1) and nearby areas from which pictures were available.

[a]	Place	Year	Age class and/or sex	Matched to our catalogue[b]	Length in m (weight in kg)	References
1	São Jacinto, Portugal	2000	Calf female	No pictures	2.20	(Marques 2005)
2	Oeiras, Portugal	2000	Juvenile female	No pictures	4.70	(Marques 2005)
3	Galicia, Spain	2001	Unknown	No pictures		CEMMA
4	Burela, Spain	2002	Adult female	NO	5.30	(Covelo et al. 2009)
5	Ouled Chaoui, Morocco	2002	Juvenile, male	No pictures	3.00	(Masski and De Stéphanis 2018)
6	50 nm offshore west of Le Verdon-sur-Mer, France	2003	Calf	No pictures		Observatoire Pelagis
7	Feunteun, France	2004	Adult female	No pictures	5.50	Observatoire Pelagis
8	Vendée, France	2004	Juvenile	No pictures	3.40	Observatoire Pelagis
9	Luaña, Spain	2004	Calf male	No pictures	2.50	CEMMA
10	Camariñas, Spain	2005	Unknown	No pictures	~4.00	(Covelo et al. 2009)
11	Azemmour, Morocco	2005	Adult male	No pictures	5.00	(Masski and De Stéphanis 2018)
12[c]	Algeciras, Spain	2006	Adult female	YES	5.70	(Guinet et al. 2007)
13	Ilhavo, Portugal	2008	Adult female	No pictures	5.00	Ferreira, pers. comm.
14	Oléron, France	2009	Calf	YES	2.45 (183)	Observatoire Pelagis
15[d]	Lauwersoog, Netherlands	2010	Juvenile	NO	3.5	(Spiegl et al. 2019) and Stranding.nl
16	Sines, Portugal	2015	Juvenile female	YES	3.40	Ferreira, pers. comm.
17	Arguin, France	2015	Unknown	No pictures	4.00	Observatoire Pelagis
18	Tahadart, Morocco	2015	Female	No pictures	5.53	(INRH 2018, 2021)
19	Viana do Castelo, Portugal	2018	Juvenile male	No pictures	4.40	Ferreira, pers. comm.
20	Sahel, Morocco	2021	Adult male	No pictures	6.50	(Kaddouri et al. 2023)
21		2016–2021		No pictures		(Kaddouri et al. 2023)
22	Lagos, Portugal	2022	Adult female	YES	5.17 (1680)	Marçalo, pers. comm.
23	Tanger, Morocco	2022	Calf female	No pictures	2.63	García de los Ríos, pers. comm.
24	River Seine, France	2022	Juvenile female	NO	4.26 (1100)	Observatoire Pelagis
25	Cadzand, Netherlands	2022	Adult female	YES	5.48 (2000)	Ijsseldijk, pers. comm.
26	De Panne, Belgium	2023	Adult male	NO	6.08	Haelters, per. comm.

[a]The numbers correspond to the locations in Figure 1. For strandings before 2000, refer to Table S1.
[b]When good quality picture(s) were available, these were compared to the photo-ID catalogue of the Iberian killer whales and a match was either found (YES), or not (NO). Alternatively, (No pictures) indicates that no good-quality pictures of their distinguishing features—such as the edge of the dorsal fin, eye patch, or saddle patch—were available or that the carcass was too decomposed.
[c]Animal was identified as Vega in Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016 but is not in the catalogue created for this study as no picture was available.
[d]Animal found sick and later transferred to a zoological facility in Spain.

From 2011 to 2023, the average proportion of adult individuals within the Iberian subpopulation was 0.61 (min = 0.53; max = 0.67) (Figure 4).

From the stranding records, lengths of females ranged from 2.2 m for a calf to a maximum of 5.8 m for adults (Tables 1 and S1). Lengths of males ranged from 2 m for a calf to a maximum

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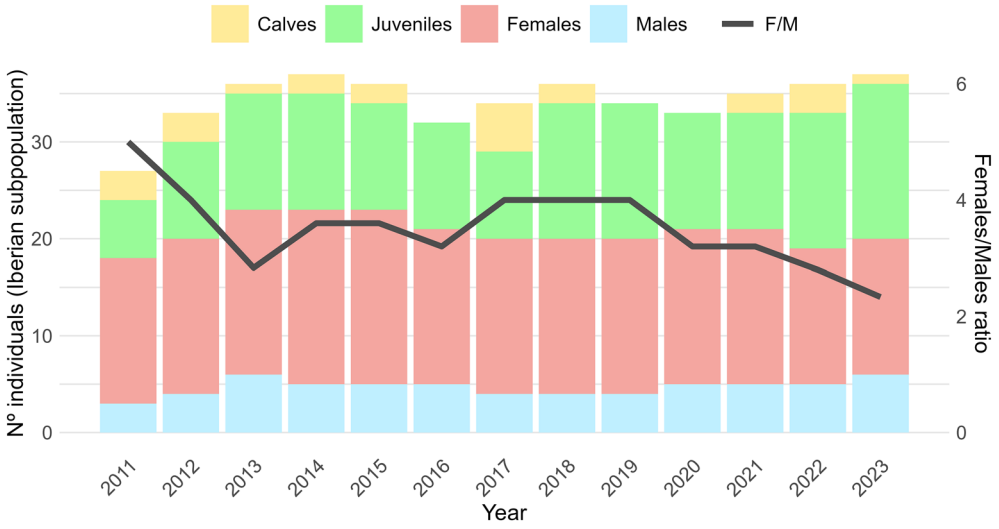


FIGURE 4 | Evolution of individuals by age class and sex. Left axis: Abundance count of individuals by age class and sex (in the case of adults) for the killer whale subpopulation of the Iberian Peninsula. Right axis: Ratio of adult females to adult males.

of 6.5 m for adults (Tables 1 and S1). Individuals with unknown sex ranged from 3 to 6.3 m (Tables 1 and S1). A higher number of strandings was registered in 2000–2009 than in the following decades (Figure S2).

3.4 | Calving Season

Calving was mostly concentrated between summer and fall, although births could occur at other times of the year. As far as we know, an orca giving birth in the wild has never been witnessed in the Iberian Peninsula. The exact dates of birth of the majority of calves born during the study were not known; the information available was that they were born sometime between consecutive whale-watching seasons (spring–summer). In some cases, however, newborn calves were first observed within a short period of time after a previous encounter with their pregnant mother, making the estimated date of birth more precise. The month of birth was estimated for three calves (2 calves in July and 1 calf in August, the latter stranded in France in December 2009, with 245 cm in length, Table 1) and the birth of four others could be assigned to a two-month period (1 calf April–May, 2 calves September–October, 1 calf December–January). Additionally, four calves of few months of age stranded along their known distributional range (Figure S1): in Peñarrubia, Spain on February 1988 with remains of the umbilical cord and unerupted teeth (male of 211 cm in length; Table S1); in Saint Palais, France on January 1994 (male of 276 cm in length; Table S1); in Muxia, Spain on May 1995 (male of about 200 cm in length; Table S1); in São Jacinto, Portugal on January 2000 (female of 220 cm in length; Table 1); in Luaña, Spain on April 2004 (male of 250 cm in length; Table 1); and in Tanger, Morocco on March 2022 (female of 263 cm in length; Table 1). Finally, a stranded female in northern Spain in April 1991 was pregnant with a fetus of 235 cm (Table S1).

3.5 | Survival and Reproductive Rates

The GOF results for the model with five reproductive states showed no overall overdispersion of the data ($\chi^2 = 20.727$, df = 38, $p = 0.99$). All the models tested are presented in Table S2. Since the two best models were within 2 ΔQAICc of each other (Table S2), model averaging was applied to models 1 and 2. Both models estimated all parameters as constant and had a similar recapture probability for non-breeding females and males. However, model 1 had a similar survival rate for breeding and non-breeding females, and model 2 estimated them separately.

Parameter estimates of the model averaging are shown in Table 2. The recapture probability was high for all reproductive states, generally above 80%. The survival rates were much lower for calves (0.838, SE: 0.083; 95% CI: 0.609–0.945) than the other reproductive states. Adult males had lower survival rates (0.933, SE: 0.038; 95% CI: 0.808–0.978) than juveniles (0.950, SE: 0.020; 95% CI: 0.893–0.978) and adult females (0.966, SE: 0.015; 95% CI: 0.923–0.986). Since all breeding females also have a non-breeding status in their capture history, the survival rate for non-breeding females (Table 2) represents the one for all adult females. Transition rates from juveniles to both male (3.3%) and female (2.4%) adults were very low. The transition between breeding states for the females indicates a low annual reproduction rate, with only 12.1% of sexually mature females giving birth per year. The inter-birth-interval (IBI) was therefore estimated at 8.3 years (95% CI: 4.1–18.3 years).

The model based only on observed reproductive females between 2011 and 2023 also showed no data dispersion ($\chi^2 = 4.626$, df = 13, $p = 0.98$). For this model, a total of 11 observed reproductive females were used, while 8 adult females that were never observed with a calf between 2011 and 2023 were excluded. The best model was $\{S^{nbF}=S^{bF}(.)\ p^{nbF}=p^{bF}(.)\ \psi^{nbFbF}(.)\}$ (model 1 in Table S3). The estimated reproductive rate was almost twice that of the previous model at 0.218 (SE: 0.046; 95% CI: 0.141–0.321), which corresponds to an IBI of 4.6 years (95% CI: 3.1–7.1 years). In this second model, the survival rate of reproductive females was 0.972 (SE: 0.017; 95% CI: 0.914–0.991), and their recapture probability was high at 0.935 (SE: 0.025; 95% CI: 0.866–0.970).

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TABLE 2 | Demographic parameter estimates for five reproductive states (C = newborn calf, J = juvenile, bF = breeding adult female, nbF = non-breeding adult female, M = adult male) and for all the individuals.

Parameter	State	Estimate	SE	LCI	UCI
Survival rate	Calf	0.838	0.083	0.609	0.945
	Juvenile	0.950	0.020	0.893	0.978
	Female (bF)	0.972	0.034	0.751	0.997
	Female (nbF)	0.966	0.015	0.921	0.986
	Male	0.933	0.038	0.808	0.978
	All	0.948	0.016	0.906	0.972
Recapture probability	Juvenile	0.898	0.029	0.826	0.943
	Female (bF)	0.853	0.232	0.134	0.995
	Male and Female (nbF)	0.799	0.035	0.722	0.859
Transition rate	J to bF	0.024	0.014	0.008	0.072
	J to M	0.033	0.016	0.012	0.084
	nbF to bF	0.120	0.047	0.054	0.246

Abbreviations: LCI = lower 95% confidence interval, SE = standard error, and UCI = upper 95% confidence interval.

The GOF test of the recruitment and population growth rate models showed transience and trap-dependence in the dataset ($\chi^2 = 49.755$, df = 27, $p = 0.005$); therefore, the models' variances were adjusted with a c-hat of 1.843. The overall annual recruitment (f) was 0.056 (SE: 0.018; 95% CI: 0.030–0.103) and the population growth rate (λ) of the Iberian killer whales was estimated close to null at 1.005 (SE: 0.021; 95% CI: 0.964–1.045) between 2011 and 2023.

4 | Discussion

The dataset used in this study is the result of a large-scale cooperation and information sharing between many different contributors. Over 40 different organizations and individuals have contributed photographs and stranding records to this study, allowing us to evaluate and update the demographic parameters of the endangered Iberian killer whale subpopulation. In particular, the high recapture probability found in mark-recapture models shows that the use of opportunistic data for a small subpopulation of identifiable animals can be suitable for this type of multistate models. Additionally, since Iberian killer whales use several distant restricted areas more regularly, their monitoring is logistically challenging and, therefore, until neighboring countries come up with a joint monitoring strategy over their range, the use of quality data from opportunistic encounters along the Iberian coast might yield a larger dataset than a monitoring plan from a single area, such as the Strait of Gibraltar. Since this study focused on demographic parameters of the subpopulation, we did not analyze social relationships and therefore could not directly compare our results with the demographic parameters of the social groups previously described in Esteban, Verborgh, Gauffier, Giménez, Foote, et al. (2016) and Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. (2016). However, in our study, we were also able to identify five communities in 2011, and an additional community was observed since 2012.

Despite the demonstrated usefulness of our dataset for mark-recapture models and community identification, the opportunistic nature of the data and the uneven effort over the study areas during the whole period limit the potential of this study. In particular, with multistate robust design models that require several sampling occasions per year, we could have yielded abundance estimates for each reproductive class (Verborgh et al. 2021). Collecting useful data for research, conservation, and management requires a long-term strategy. Although the Spanish Conservation Plan for this endangered subpopulation was approved in 2017 (BOE-A-2017-5474 2017), the Spanish administrative process to implement its monitoring only started in 2021 (Ministerio para la Transición Ecológica y el Reto Demográfico 2021). Further studies, such as assessing the genetic relatedness of the visitor individuals, are also important to evaluate the status of this subpopulation.

A previous study estimated survival rates for Iberian killer whales between 1999 and 2011 (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016), where none of the calves survived their first year of life after 2005, related to the decrease in tuna drop-line fishery catches in these years. Although the survival rate was lower for calves than for older animals between 2011 and 2023, which is also the case for other cetacean populations (Olesiuk et al. 2005; Verborgh et al. 2021), it has greatly increased from the null value of 2005–2011. The previous survival rate for the whole population was higher, estimated at 0.959 (95% CI: 0.892–0.985; Ouled-Cheikh et al. 2023) for the period 1999 until 2011. Survival rates for the other reproductive states were not directly comparable with the previous study, as Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. (2016) estimated these parameters separately for individuals depredating or not on the tuna fishery and together for males and females. Our results are then lower than the 1999–2011 survival estimates for depredating juvenile (0.966; SE = 0.024) and adult whales (0.991; SE = 0.011) but higher than those of non-depredating adult whales. These are low survival rates given the expected long lifespan of the species and correspond to a general decrease of adult individuals in the last 10 years of the study, especially for females.

Regarding female reproductive rates, our inter-birth interval including all females was similar to the 7 years described in the previous period (1999–2011) for viable calves only (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016). In our study, females that reached menopause, and therefore did not reproduce in their older years (estimated at 48 years in the Pacific Ocean in Olesiuk et al. 1990), were not excluded from this model, which could lead to an underestimated female reproductive rate. However, other factors could also have contributed to this low value. In particular, mortality during pregnancy or very early after birth could have been missed, especially due to the opportunistic nature of our data, and therefore not be included in the reproductive rate. The second model, using only females

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that were observed at least once with a calf between 2011 and 2023, estimated a reproductive rate twice as high as the previous model, and similar to the reproductive rate estimated for the period 1999–2011 for all females, whether their calf survived or not (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016). This estimate of calving rate is similar to other populations (Table S4). In this reduced model, a large proportion of adult females ($n = 8$, 42%) that were not seen with a calf during the modeling period were excluded, which explains the different results.

Most births took place in the summer and fall, showing a scattered seasonal pattern. However, the paucity of mid-winter births could also be attributable to a reduced photographic effort. It is unclear to what extent the reproductive cycle varies in other regions. An apparent synchronized calving season along the Pacific Ocean was suggested as most of the births were occurring between September and November in British Columbia (Olesiuk et al. 1990, 2005), and in Japan as indicated by the lengths of fetuses collected, assuming a 16-month gestation period (Nishiwaki and Handa 1958). Antarctic catches taken in January–March (Anderson 1982) suggested a peak in parturition in about January–April (i.e., late austral summer). The seasonality of calf sightings in British and Irish waters suggested calves were born mainly between November and February (Evans 1988).

The very low transition rates from juveniles to adults, either male or female, can be partly explained by the lack of recruitment in the population in the years 2005–2010 (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016), preceding the modeled period in this study. Another limitation is intrinsic to our definition of adult females, which relies upon being observed the first year they give birth and, as mentioned previously, for the newborn not to die before the female is observed. Although most of the births occurred during the spring and summer seasons, coinciding with the maximum whale-watching activity in the area, some births could have been missed. The latter could be particularly relevant as first-born offspring receive greater contaminant loads from their mothers than subsequent offspring (Ylitalo et al. 2001). If these two conditions are not met, then first-time mothers would still be considered as juveniles in our data. Nevertheless, the oldest juvenile in 2023 was only born in 2011. Therefore, most transitions resulted from calves born either before 2005 or after 2011. Only 4 juveniles became adult males and 3 juveniles became adult females between 2011 and 2023. Furthermore, there was only one individual, born in 2012, that became an adult male in 2023 after 12 years, and another, born in 2005, that became an adult female in 2015 after 11 years, showing the low probability of observing the transition during the 13-year modeling period.

The relatively low values of survival and recruitment estimated for 2011–2023 lead to an apparently null population growth rate (0.46%) for the Iberian subpopulation, while a 4% positive growth was observed in the previous period (1999–2011) for individuals depredating tuna fishing boats (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016). The annual population growth rate of some thriving killer whale populations is probably at its maximum, such as the southern Alaska resident population (3.5%; Matkin et al. 2014; Table S4). However, decreasing population growth rates have also been observed, as for the population in British and Irish waters (Beck et al. 2013; Table S4), or the 'southern resident' individuals in British Columbia (Department of Fisheries and Oceans Canada [DFO] 2018; Table S4). Additionally, the Iberian subpopulation showed lower survival for the younger stages than most killer whale populations (Table S4), while adult survival rates were similar to populations affected by human impacts that impact their viability, such as those inhabiting the Crozet Archipelago (Poncelet et al. 2010; Tixier et al. 2010), or even post-reproductive females (> 42 years old) in British Columbia (Krahn et al. 2002), which is concerning for the future of the subpopulation.

Of the individuals considered dead, only a small fraction was confirmed through strandings, suggesting that many deaths go undocumented. In fact, a significant portion of known strandings either involved individuals without identification pictures or were too decomposed for identification. This highlights the limitations of relying solely on stranding data to assess mortality. While strandings do offer valuable insights, they likely underestimate true mortality rates, as many animals may die at sea or strand in remote locations where they go unnoticed. In this study, only a small portion of identifiable (with pictures) recorded strandings (33%) were individuals not included in our catalogue. It is possible that these individuals were either (i) outside their typical range and not part of this subpopulation (17%; Figure S1), or (ii) stranded at an early stage of our study before all individuals were photo-identified and catalogued (16% before 2003). In either case, the majority (67%) of identifiable recovered carcasses were of individuals that stranded within their known distribution range and were part of the Iberian subpopulation. Nevertheless, stranding records helped us consider that the killer whales of the Iberian Peninsula appear to differ in body size compared to other populations (Table S5): except for Antarctic Type C whales, the Iberian killer whales are generally smaller than the average size of other killer whale populations globally. None of the stranded females reached 6 m, so most of the undetermined individuals over 6 m were more likely males (Table 1 and S1). Additional research should be conducted using size-at-age data to infer long-term growth trends, and aerial photogrammetry data may improve our ability to estimate size disparities with other populations (e.g., Fearnbach et al. 2011).

The maximum number of animals that can live in a certain area depends on the quantity of resources present in the environment and the needs of each animal. Carrying capacity is frequently determined by the availability of food (e.g., Beck et al. 2006). Prey abundance drives both mortality and reproductive success in resident killer whales in British Columbia (Ford et al. 2010; Ward 2009). The positive abundance trends in recent years of Atlantic bluefin tuna, the main prey of Iberian killer whales, have not led to improvements in the survival and reproduction of these whales. A high number of strandings were registered in their known distributional area before this study (1990–2009; Figure S2), suggesting that a higher number of individuals have been sustained in the Iberian Peninsula, and that the actual abundances of killer whales do not reflect the limit of the carrying capacity. Additionally, other factors can limit the number of animals in an area. Genetic factors such as inbreeding depression can also affect population dynamics. Southern residents, which are currently endangered despite nearly 50 years of conservation efforts, exhibit strong inbreeding depression for survival (Kardos et al. 2023). The requirement for space can also be a factor, and the Strait of Gibraltar, which seems

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to still be their main foraging area at least during part of the year, is a restricted area.

Furthermore, other threats may be preventing the recovery of the Iberian subpopulation of killer whales. Individuals sampled in the Strait of Gibraltar had very high levels of PCBs (Jepson et al. 2016), above the highest PCB toxicity threshold published for marine mammals based on marked reproductive impairment in seals (Helle et al. 1976). In the Strait of Gibraltar, killer whales have been bearing abrasion marks that could come from repeated contact with fishing lines (Herr et al. 2020), since they started to depredate tuna from the Spanish and Moroccan drop-line fisheries (Esteban, Verborgh, Gauffier, Giménez, Guinet, et al. 2016). However, some whales have recently exhibited more concerning injuries likely caused by their close proximity with fisheries since at least August 2015, when a female was observed with a severed right flipper together with a cut at the caudal base of the dorsal fin (Herr et al. 2020). This individual was subsequently observed every year with healed wounds until 2022 when it stranded in Portugal (Marçalo pers. comm., Table 1). Fishing lines have been observed attached to animals in the Strait of Gibraltar (Herr et al. 2020) and in Algarve in 2023 (Abrantes pers. comm.). Furthermore, four young whales were reported entangled in fishing buoys in Portugal. Three of these were rescued by whale-watching crews in Algarve, one juvenile in August 2018 (Abrantes and Magalhães pers. comm.), the second juvenile in May 2019 (Magalhães pers. comm.), and a calf in June 2019 (Magalhães pers. comm.). Another juvenile was rescued from an entanglement in a fishing buoy in November 2022 near Sesimbra (Fishermen from Sesimbra 2022). Other types of injuries have also been observed, such as the retained tag dart on a juvenile observed from 2011 until 2013. Since 2016, a drop-line fishing company has been using a new fishing technique in the Strait of Gibraltar that includes a surface line with lures (the "GreenStick"; JC Mackintosh 2022a). The next year, the same company started applying a small electric current to stun the tuna caught on the line instead of getting it tired before lifting it to the surface (JC Mackintosh 2022b). To the extent of our knowledge, the potential risks of these new techniques for depredating killer whales have not been assessed. Additionally, since the interactions with recreational boats started in 2020 (Esteban et al. 2022), two animals identified as interacting animals were observed with injuries. One was a juvenile reported with deep lacerating wounds of unknown origin (Esteban et al. 2023), and the other one was an adult female observed in August 2020 with five parallel wounds and a dorsal abrasion (Hanninger et al. 2023). The impact of each of these threats both separately and cumulatively on Iberian killer whales is not fully understood, but they might have impacted the survival and reproductive parameters of this critically endangered subpopulation.

5 | Conclusion

Overall, the Iberian killer whale subpopulation was stable during the updated modeled period (2011–2023), probably mainly because of the recovery of bluefin tuna. However, their low abundance still qualifies for the critically endangered category of the IUCN Red List, and the continuous decrease of mature individuals is concerning. The future of the subpopulation will be highly dependent on the recruitment of juvenile females into effective reproductive females and the survival of their calves for the subpopulation to grow above non-threatened levels. Only long-term monitoring of the subpopulation, together with international collaborations, will allow assessment of the upcoming state of this subpopulation in which each individual has a key role to play for their future.

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Acknowledgments

We would like to especially thank the people and organizations that provided opportunistic pictures such as the scientific team of Tumares Tarifa S.L., Dream Wave Algarve, José Martínez Cedeira from CEMMA, Jon Hidalgo Mugica, Javier Elorriaga, Yeray Seminario, Rita Leitão, Rotas da Sal, Raül Luna Escarda, Pedro Narra, Fabio Pais from SOMAR, Vasco Varanda, Maite Louzao from AZTI, Catarina Gomes, Ricardo Tomaz, and Andrew Watson. Several people contributed with information and pictures of stranded animals, so we would like to thank Marisa Ferreira from Sociedade Portuguesa de Vida Selvagem, Ana Marçalo from Rede de Arrojamentos do Algarve, Lonneke IJsseldijk from Universiteit Utrecht, and Álvaro García de los Rios from Centro de Estudio y Conservación de Animales Marinos. We also wish to warmly thank all members of the French stranding scheme (RNE) and all members of Pelagis observatory for their continuous effort in collecting data on stranded cetaceans. We thank the reviewers and the editor for their constructive and insightful comments, which helped to improve the clarity and quality of the manuscript. The work of A.L. is supported by the CESAM by FCT/MCTES (UIDP/50017/2020+UIDB/50017/2020+LA/P/0094/2020), through national funds and by national funds (OE), through FCT-I.P., in the scope of the framework contract foreseen in the numbers 4, 5, and 6 of the article 23 of the Decree-Law 57/2016 of August 29, changed by Law 57/2017 of July 19. All whale watching companies were authorized to carry out their activities by the competent administration, the Instituto para a Conservação da Natureza e a Floresta (ICNF) in Portugal and

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the Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO) in Spain. This work was partially funded by the project FriendSHIP: Orca with funds from the Fundación Santander (https://www.orcaiberica.org/; App: GT ORCAS).

This paper is dedicated to the memory of David Alarcón, who passed away in 2022. Throughout his life as an environmental journalist and wildlife photographer, he played a crucial role in the study and conservation of the subpopulation of Iberian killer whales by taking part in numerous research surveys and leading public outreach initiatives.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1]The subpopulation was also assessed as vulnerable in the Spanish Catalogue of Endangered Species in 2011 (Gobierno de España 2011), but in France, it is still categorized as Data Deficient (IUCN France 2017). Killer whales are also protected by the Moroccan Law on the protection of species of wild flora and fauna and the control of their trade (Loi 29–05 2011), as any species belonging to Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). These countries are also Parties to the Agreement on the Conservation of Cetaceans of the Black Sea, Mediterranean Sea and contiguous Atlantic Area (ACCOBAMS), and therefore "shall take co-ordinated measures to achieve and maintain a favorable conservation status for cetaceans" and "shall prohibit and take all necessary measures to eliminate, where this is not already done, any deliberate taking of cetaceans" (article II.1 of the Agreement), including injuring or harassing and attempting to engage in any such conduct. Additionally, Resolution 8.12 urges Parties to implement measures to address the threats posed to the Critically Endangered subpopulation of Strait of Gibraltar/Iberian killer whales, to update their national Red Lists according to this status and to take appropriate conservation actions for this subpopulation (https://accobams.org/wp-content/uploads/2023/01/MOP8.Doc31_Annex13_Res8.12.pdf).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Strandings and captures of killer whales along the distributional range (Esteban et al. 2022; Pouey-Santalou et al. 2024) of the Iberian subpopulation previous to the study period (1846–1999). Numbers are the same ones as in Table S1. **Figure S2:** Number of killer whale strandings along the distributional range (Figure S1) of the Iberian subpopulation. **Figure S3:**. Photographic effort before 2011, notice that values are not in thousands. **Table S1:** Compilation of killer whales stranded or captured (1846–1999) along their known distributional range (Figure S1), No catalogue was available for this period, so no matching was done. **Table S2:** Results of the multistate models with the modeling step, model number, the models tested, the value of the QAICc, the QAICc difference with the best model (ΔQAICc), the weight of each model (QAICc weight), the number of parameters used in the model (NP), and deviance (QDev). Parameters noted (.) were constant and (t) were time-dependent. **Table S3:** Results of the multistate models with only reproductive females showing the modeling step, model number, the models tested, the value of the QAICc, the QAICc difference with the best model (ΔQAICc), the weight of each model (QAICc weight), the number of parameters used in the model (NP), and deviance (QDev). Parameters noted (.) were constant and (t) were time-dependent. **Table S4:** Estimates of life-history parameters of killer whales from the Iberian Peninsula compared to other populations. Some estimates were obtained separately for breeding females (bF) and non-breeding females (nbF). Estimates ± SE and 95% confidence interval between brackets. * indicates SD instead of SE. **Table S5:** Summary of length of killer whales from the Iberian Peninsula compared to other populations. Length measurements were obtained as ranges (min-max) or averages (underlined), other types were clarified with text and standard deviation are between brackets. When the estimates for the different sex were estimated only for adult individuals it is indicated in italics.

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